

Greatest Common Factor

Lesson 1-2

Name: Type your name **Date:** Today's date **Class:** Class period



TODAY'S OBJECTIVES

Content Objective: I can find the greatest common factor (GCF) of two numbers by listing or comparing their factors.

Language Objective: I can explain how I found the GCF using the words factor, common factor, divisible, and greatest common factor.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Factor

Greatest Common Factor

Common factor

Divisible

Prime factorization



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

A number that divides evenly into another number with no remainder is a

2

— The biggest number that divides two or more numbers evenly.

3

A factor that two or more numbers share is a factor.

4

A number that can be divided by another with no remainder is

 by that number.

5

— Writing a number as prime numbers multiplied together. It helps you find the GCF.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

How did you find the factors that 24 and 36 share, and how did you decide which one is the GCF?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

☐

Factor — A number that divides another number with nothing left over.

Factor — Un número que divide a otro sin que sobre nada.

☐

Greatest Common Factor — The biggest number that divides two or more numbers evenly.

Máximo común divisor — El número más grande que divide a dos o más números sin que sobre nada.

☐

Common factor — A number that divides two or more numbers evenly.

Factor común — Un número que divide a dos o más números sin que sobre nada.

☐

Divisible — A number you can divide evenly, with nothing left over.

Divisible — Un número que se puede dividir sin que sobre nada.

☐

Prime factorization — Writing a number as prime numbers multiplied together. It helps you find the GCF.

Factorización prima — Escribir un número como números primos multiplicados. Ayuda a encontrar el MCD.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

The common factors are ____.

Los factores comunes son ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: The number of pods must divide BOTH 24 and 36 evenly.

Evidence: Students see the number of pods has to be a common factor of 24 and 36, so it must divide both with nothing left over, ruling out numbers that only divide one of them.

Reasoning: This evidence connects the definition of factor to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The common factors are ____.**
Los factores comunes son ____.
- **The greatest common factor is ____ because ____.**
El máximo común divisor es ____ porque ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
Mi afirmación es ____.
- **I know because ____.**
Lo sé porque ____.
- **So ____.**
Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.

- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.

- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.

- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.

- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.